The yen, dollar and Big Mac Index

Understanding prices and exchange rates through one product

What is the Big Mac Index?

The Economist devised this indicator to compare Big Mac prices across economies. It offers a simple way to think about purchasing power and exchange rates.

Find the rate that equalizes prices

Japanese price in yen ÷ US price in dollars. This gives an exchange rate that would equalize the price of one product. Compare it with the market rate.

How to read this guide

The following pages use hypothetical numbers. They are teaching examples, not observed menu prices, market exchange rates or survey results.

Converting prices into yen

Hypothetical example: these are not observed prices or exchange rates

Assume three numbers

Japanese price: JPY 600
US price: USD 6
Exchange rate: JPY 150 per USD

Convert the US price

USD 6 × JPY 150/USD = JPY 900.
Compared with JPY 600 in Japan, 900 ÷ 600 = 1.5 times the price.

Do not attribute the whole gap to currency

In this example, the US price is higher in yen terms. Wages, rents, taxes and other local factors can also affect the gap.

The rate that equalizes prices

Using the same hypothetical assumptions

Simple purchasing power parity

JPY 600 ÷ USD 6 = JPY 100 per USD.
At this rate, the assumed prices are equal.

Compare with the assumed market rate

(100 ÷ 150 - 1) × 100 ~ -33.3%.
By this single-product measure, the yen would be about 33.3% undervalued against the dollar.

JPY 100/USD is not a fair-value claim

This illustrates the calculation. It is neither an estimate of current yen undervaluation nor a forecast that the exchange rate will move to 100.

What the index shows - and its limits

One product is not a complete measure of living standards

Local costs are part of the price

Wages, rents, taxes, logistics and competition affect menu prices. A simple price comparison cannot fully adjust for these differences.

Look beyond a single product

Income, housing and healthcare also matter. A cheap Big Mac alone does not prove that a country is rich or poor.

Separate comparison from prediction

Use the index as an introduction to purchasing power, not as an answer to short-term currency or stock-market forecasts.

Concept reference

The Economist: The Big Mac index.
Original explanation, diagrams and hypothetical calculations by TODOROKI.

https://github.com/TheEconomist/big-mac-data